

Vehicle Management System

Feature Guide - Barangay Emergency Response Vehicle Management

Group 4 | What the system does today, and what is planned

Overview

The Vehicle Management System (VMS) is a web-based desktop system that helps the barangay organize, monitor, and maintain its emergency response vehicles. It gives Admin, Custodian, and Maintenance Personnel a shared, accountable record of every vehicle - its condition, reported issues, repairs, schedules, and readiness to respond. This guide lists what the system does today and the enhancements planned to make it capture reality more completely.

Part A - What the System Can Do Today

Platform & Access

- Web desktop application with role-based access for Admin, Custodian, and Maintenance Personnel; one account may hold several roles.
- Secure login (rate-limited), logout, password change, and active/inactive account enforcement.
- Cloud file and image uploads - issue photos, repair attachments, and vehicle photos.
- User account management: create accounts, activate and deactivate users.

Vehicle Records

- Full vehicle records with create, edit, archive/restore, and decommission.
- Vehicle categories/types and locations/hubs (with map coordinates).
- Vehicle statuses: Available, Under Maintenance, Inactive, Decommissioned.
- Ready-to-Respond readiness: readiness checks with 24-hour freshness, readiness states, a Ready-to-Respond column and stat cards, plus add/remove of readiness-check items.
- Vehicle reliability view.

Condition Monitoring

- Periodic custodian condition checks recording each vehicle's physical condition over time.

Issue Reports

- Issue report records with photo evidence, severity level, and status tracking.
- Duplicate / open-ticket warning before filing a new issue on a vehicle.
- One-click Create Ticket directly from an issue report.

Maintenance Records

- Standalone maintenance records with a Verify (Custodian) then Confirm (Admin) lifecycle.
- Progress states: Assigned, Under Repair, For Verification, Completed.
- Maintenance Personnel can self-file a record; Admins are notified; a new issue report can be created inline from the form.

Maintenance Schedules (Preventive)

- Preventive schedules that, on completion, auto-generate the next recurrence at a chosen interval (including a custom number of months) while avoiding double-booking.
- Default "Scheduled only" view with pagination.

Ticket Workflow (Request -> Validate -> Approval engine)

- Create (Admin) -> assign Custodian -> Inspect (Custodian; result Needs Maintenance or No Issues, creating

sub-issues) -> Assign mechanic per sub-issue -> Log repairs (Mechanic) -> Verify (Custodian functional-test checklist + attestation) -> Confirm (Admin) -> Done.

- Add a sub-issue mid-ticket; reassign a mechanic; reopen/unconfirm a Done sub-issue; defer a sub-issue (opens a breadcrumb issue report).
- Close, cancel/uncancel, or delete a ticket; assignment-based verification (only the assigned custodian may verify).
- Ticket archive audit log with the ability to reopen an accidentally deleted ticket.
- Recurrence count - how many times the same issue recurred in the last 90 days.

Dashboards & Oversight

- Role-specific dashboards for Admin, Custodian, and Maintenance.
- Emergency Readiness dashboard card with verified-ready as the headline number.
- Reports (vehicle, issue, maintenance, history), activity logs, and a per-vehicle history timeline.
- Color-coded notifications by type, with read/unread tracking.
- Work Tracker - Custodian and Maintenance see the vehicles they have worked on and the outcomes, with filters.

Part B - Planned Enhancements (13)

These build on the existing system and stay within the project scope. Each turns a real-world situation the system currently drops into captured, usable data.

Flexible Ticket Entry (enter the workflow at the right phase)

- 1. Pre-diagnosed tickets (Mode B) - when the problem is already known, the ticket skips the redundant inspection and is born ready for a mechanic.
- 2. Create ticket from a condition check (Mode C) - a "Needs Repair" condition check gets a one-click Create Ticket that carries its findings in, pre-diagnosed.

Mode A (full inspection) remains today's flow, as an opt-in for genuinely unknown symptoms. Mode D (emergency field repair) is already served by the mechanic self-file standalone maintenance record - one governance decision remains (full confirm vs. acknowledge-and-close).

Truthful Data Capture

- 3. Real down/up times (backdating) - log when the vehicle actually went down and returned, so downtime figures are accurate.
- 4. "On Hold - Awaiting Parts" state - separates a slow mechanic from a slow supplier.
- 5. External-shop repair record - vendor, cost, warranty-until date, and receipt photo for third-party repairs.
- 6. "Reported on behalf of [driver]" field - a text box that keeps the real eyewitness on the record (no driver login needed).
- 7. Duplicate-report warning - catch two people reporting the same issue before two tickets open.

Learning from History

- 8. Standardized issue catalog - pick a fault type from a list instead of free text, so recurrence and duplicate detection work reliably. (Build first - items 9 and 10 depend on it.)
- 9. "It broke again" recurrence link - connect a re-break to the original closed ticket to surface repeat failures.
- 10. Lifetime cost per vehicle - total every repair and flag when a vehicle is consuming too much of its value (decommission signal).

Readiness & Accountability

- 11. Overdue-PM flag - a missed or overdue scheduled maintenance actively warns instead of sitting silent.
- 12. Fleet-level readiness summary - one figure and alert: how many emergency vehicles are ready right now, turning red below a set minimum.
- 13. Edit-trail on records - who changed what and when on sensitive/closed records (serves the "logs for

accountability" must-have; essential once backdating is allowed).

Build order & open decision

- Dependency: build #8 (standardized issue catalog) before #9 and #10 - recurrence and cost-by-fault rely on standardized fault types.
- Open decision (Mode D): when a mechanic self-files an emergency repair that is already fixed and back in service, does an Admin do a full confirm, or simply acknowledge-and-close? Recommended: acknowledge-and-close, still logged - fast but auditable.

Appendix - Consciously Out of Scope

These are real needs deliberately left out because they fall outside the project scope (web desktop only; roles limited to Admin/Custodian/Maintenance; no operations/dispatch, inventory, or mileage). They are noted so the boundary is a conscious choice.

- Offline / mobile field capture - scope is a web desktop platform.
- Dispatch-anyway override and post-emergency-use checks - require operations/deployment tracking.
- Consumables/stock readiness (oxygen, water, fuel) and parts cannibalization - require an inventory domain.
- Shift-roster handoff - requires an on-duty roster concept.

Of these, consumables readiness is the most emergency-relevant candidate should the scope ever be widened.